⚡ ADHD MODE — LIVING REFERENCE DOC

ENCLAVE

Your AI. Your Data. Your Space.

// Last updated: 2026-03-28 // Repo: github.com/mtldev-dotcom/clawhost

// Production: nestai.nickybruno.com // Dokploy: 35.202.32.236:3000

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🧠 01 — WHAT IS THIS

Multi-tenant SaaS that gives every user a **private, hosted AI co-worker**. One subscription → dedicated AI agent with a custom subdomain, 25+ messaging channels, shared file workspace, deep third-party integrations, and a memory that **never forgets anything**.

> **TL;DR:** Think "Google Drive + AI assistant + zero data exposure" — set up in 60 seconds, no DevOps required. Built on OpenClaw (337k ⭐ on GitHub).

🎯 02 — THE PITCH

PROBLEMS WE SOLVE

- ✗ **AI forgets everything** — ChatGPT/Claude reset every session. After 1 year of daily use, they know nothing more than day one.
- ✗ **No true privacy** — Every hosted AI platform can read your conversations. We physically cannot.
- ✗ **No shared workspace** — AI integrations are isolated silos. No single agent spans all your tools.
- ✗ **Setup is a barrier** — OpenClaw has 337k stars but requires terminal + Docker + Node. Fewer than 5% of people who'd benefit can self-host it.

OUR ANSWER

- ✓ Private container per user — encryption key we never store
- ✓ DAG-based memory that stores every message, forever
- ✓ Shared file workspace user + agent both read and write
- ✓ 25+ channels, 40+ integrations, 30+ pre-built skills
- ✓ Managed hosting — zero DevOps for the user

VS. COMPETITION

PRODUCT	FATAL WEAKNESS VS. ENCLAVE
ChatGPT / Claude.ai	Shared infra, session resets, provider reads your data, no workspace
Zapier AI	Automation tool, not a co-worker, no memory, no private hosting
Notion AI	Locked inside Notion, can't act on other tools, no conversation memory
Personal.ai / Mem.ai	Memory-only, no agent execution, no private hosting
Self-hosted OpenClaw	Requires DevOps, no managed workspace, no skills marketplace
✦ Enclave	Private instance + never forgets + full workspace + 25 channels + managed

🏛️ 03 — FIVE PILLARS

PILLAR	PROMISE
🔐 Private by Architecture	Isolated container per user. 256-bit key injected at provision — we never store it. We physically cannot read your data.
🧠 Never Forgets	DAG-based full message persistence + semantic vector search + entity extraction. Ask about anything from any conversation, ever.
📁 Shared Workspace	A file system both user and agent can read, write, and organize. The AI has a desk in your office.
💬 Works Everywhere	25+ messaging channels natively. Agent follows you wherever you work.
🔌 Extensible by Design	Skills marketplace, MCP integrations, cron automation, webhooks, multi-agent teams.

📁 04 — THE SHARED WORKSPACE (VIRTUAL OFFICE)

> **The Analogy:** Google Drive exists so you can store files. Your Enclave Workspace exists so you *and your AI* can store, organize, and act on files together. The AI is not a visitor — it's a co-owner with its own desk.

ENCLAVE DRIVE

FEATURE	WHAT IT DOES
Upload & Organize	PDF, DOCX, images, spreadsheets, code — agent reads and indexes automatically
Agent Can Write	Agent creates, edits, and saves files. Reports, research notes, spreadsheets appear in your file list.
Full-Text Search	Search across every file by content. "Find the

	contract with the Montreal clause" returns the right file.
Version History	Every agent change is versioned. Review and roll back.
Folders & Projects	Organize by client or project. Agent knows which context applies.
Large File Handling	Files over 25k tokens stored separately, summarized for context. Full content retrievable on demand.
Shared Notes	Markdown notes layer. User and agent both create notes. Cross-referenced with conversations.

KNOWLEDGE BASE

- **Personal SOP Library** — "Always CC Sarah on client emails"
- **Tone & Style Guide** — writing preferences applied to all drafts
- **Key Contacts** — names, roles, relationship notes
- **Project Briefs** — background docs agent always has access to
- **Reference Materials** — policies, rate cards, templates

MEMORY DASHBOARD

- **Memory Timeline** — chronological view of everything agent learned
- **Entity Browser** — browse contacts, projects, appointments, preferences
- **Edit & Delete** — correct wrong memories, delete sensitive info (GDPR)
- **Memory Export** — full download. You own your data.

ACTIVITY LOG

Every agent action logged with timestamp, tool used, and outcome. Full audit trail. No hidden actions.

🧬 05 — MEMORY ENGINE (NEVER FORGET)

Layer 1 — Verbatim Recall: `lossless-claw`

Replaces OpenClaw's sliding-window truncation with a DAG summarization system. Every message stored in SQLite. Nothing ever deleted.

- **lcm_grep** — full-text + regex search across all messages and summaries
- **lcm_expand_query** — deep recall via sub-agent delegation, drills down to raw messages
- **lcm_describe** — inspect specific DAG nodes or stored files by ID
- **Depth-aware summaries** — leaf → condensed → arc-level. Nothing discarded.
- **Large file interception** — files >25k tokens stored separately, retrievable on demand

> **Practical demo:** March 2027 — "What was that contractor's name from January 2026?" → Agent searches DAG, expands summary back to original message, returns exact answer. Not an approximation.

> ⚠ **Dependency:** lossless-claw requires OpenClaw PR #22201 (not yet merged). Must fork + patch until upstream merges. Prioritize this.

Layer 2 — Semantic Recall: Qdrant

Self-hosted vector DB sidecar. Meaning-based retrieval — "Find that Montreal renovation project" works even if those exact words were never used.

- Runs entirely in-container — no data leaves the instance
- Cross-document search: conversations + workspace + knowledge base simultaneously
- MCP tool bridge — OpenClaw queries it natively mid-conversation

Layer 3 — Structured Memory: Memo

Self-hosted AI memory layer. Auto-extracts entities from conversations into a personal knowledge graph.

MEMORY TYPE	EXAMPLE
People & Contacts	"Sarah is my accountant, +1-514-xxx-xxxx, uses QuickBooks Online"
Projects & Clients	"Project Atlas = bakery client e-commerce, deadline Q4 2026, budget $15k"

Preferences	"Nick prefers formal French emails, no bullet points, always BCC himself"
Appointments	"Dentist every 6 months, last visit March 2026 with Dr. Leblanc"
Recurring Tasks	"Invoice Marc on the 1st of every month, $3,500 + tax"
Goals & Commitments	"Committed to shipping v2 before end of April, blocked on API design"

PRIVACY ARCHITECTURE

MECHANISM	HOW IT WORKS
Key Generation	256-bit key generated at provision, injected as env var into container — never written to our DB
Data Isolation	SQLite + Qdrant + Mem0 + workspace files all encrypted at rest with user's key
Zero Operator Access	We store: email, subscription status, instance URL. Nothing else. Subpoena returns nothing.
Backup Ownership	Optional nightly export to user-owned S3/R2 bucket. Their key, their bucket.
GDPR Deletion	Delete account → destroys container + volume + all keys. Cryptographically irreversible.

💬 06 — CHANNELS (25+)

CHANNEL	STATUS	NOTES
Telegram	Live	Primary default. DM + group, pairing code security, webhook mode.
WhatsApp	Live	Baileys-based.

		Personal + business accounts.
Discord	Phase 1	Bot token, DM + server channels, slash commands.
Slack	Phase 1	Bolt integration, workspace + DM routing.
Google Chat	Phase 1	Google Chat API, space + DM.
Web Chat (Dashboard)	Phase 1	Browser-based inside the Enclave dashboard.
iMessage / BlueBubbles	Phase 2	BlueBubbles server recommended.
Signal	Phase 2	signal-cli. Maximum privacy.
Microsoft Teams	Phase 2	Bot Framework integration.
Matrix	Phase 2	Decentralized privacy-first network.
Email (reply-by-email)	Phase 3	Gmail Pub/Sub webhooks. Real-time inbox.
LINE / Feishu / Mattermost	Phase 3	Asian markets + enterprise.
Nextcloud / Synology Chat	Phase 3	Self-hosted messaging platforms.

Voice (Talk Mode)	Phase 4	Voice Wake + continuous voice on Android.
WeChat / Nostr / Tlon / Twitch / Zalo / IRC	Phase 4+	Niche + decentralized + streaming markets.

🔌 07 — THIRD-PARTY INTEGRATIONS

All integrations surfaced as MCP tools — agent doesn't just *read* data, it takes *real action*.

TIER 1 — PRODUCTIVITY CORE (PHASE 1-2)

APP	ACCESS	AGENT CAN
Google Calendar	R/W	Schedule, check availability, set reminders, morning briefings, block focus time
Gmail / Google Mail	R/W	Triage inbox, draft replies, send with approval, Gmail Pub/Sub real-time alerts
Google Drive / Docs	R/W	Read + write docs in existing Drive, sync to workspace, co-edit
Google Sheets	R/W	Query data, update cells, generate reports, create charts
Notion	R/W	Create pages, update databases, manage wikis, build project trackers
Outlook / Microsoft 365	R/W	Microsoft equivalent of Gmail + Calendar

TIER 2 — TASKS & TEAMS (PHASE 2-3)

TickTick · Todoist · Airtable · Slack · Discord · Linear · GitHub · Jira · Asana

TIER 3 — BUSINESS & FINANCE (PHASE 3-4)

Stripe · QuickBooks · FreshBooks · HubSpot · Shopify · Calendly · Zapier / Make

TIER 4 — CONTENT & MEDIA (PHASE 4)

WordPress · Ghost · Buffer · YouTube · RSS / News · Readwise · Spotify

TIER 5 — DEVELOPER & POWER USER (PHASE 4-5)

Sentry · Railway / Render / Fly.io · Cloudflare · Vercel · Tailscale · NixOS / Docker / SSH

🧩 08 — SKILLS MARKETPLACE

Pre-built, one-click workflows. OpenClaw's ClawHub already hosts 1,700+ community skills. Enclave curates + packages them + adds a premium tier.

CORE SKILLS (PRO TIER)

SKILL	WHAT IT DOES
☀️ Morning Briefing	Weather + calendar + priority emails + open tasks. Delivered to Telegram. Adapts to calendar density.
🌙 Evening Wind-Down	What was accomplished, what carries over, tomorrow's preview, overnight flags.
📋 Meeting Prep	Before any calendar event: research attendees, draft talking points, surface relevant files, prepare agenda.
📝 Meeting Notes	Transcribe audio → extract action items → assign by name → create tasks in TickTick/Jira/Todoist.
📥 Inbox Zero	Triage email: categorize, draft replies for approval, archive low-priority. On schedule or on-demand.
📊 Weekly Review	Every Sunday: pull completed tasks + calendar + metrics → write summary → post to Notion or Telegram.

RESEARCH SKILLS

SKILL	WHAT IT DOES
🔍 Lead Researcher	Company or person name → comprehensive brief: news, social presence, suggested outreach angles.
🗺️ Market Scan	GitHub + HN + Product Hunt + Reddit + X → competitive landscape, open gaps, sentiment.
📡 Competitor Monitor	Weekly scrape of competitor sites for pricing changes + new features. Changelog-style diff report.
📚 ArXiv Reader	Fetch, summarize, cross-reference academic papers. Personal research library in workspace.
💢 Pain Point Miner	Reddit/X monitoring for complaints about a product category. Weekly digest for product validation.
📈 Earnings Tracker	Monitor SEC filings + press releases for specified companies. Alert on earnings calls.

CONTENT SKILLS

SKILL	WHAT IT DOES
♻️ Content Repurposer	1 blog post → Twitter thread + LinkedIn post + email newsletter + TikTok script
✉️ Newsletter Writer	Research + draft + remember previous issues + format for Substack/Ghost. Scheduled.
🔎 SEO Blog Engine	Research keyword → multi-source data → structured draft → light editing needed.
▶️ YouTube Summarizer	Daily digest of new videos from followed channels → workspace notes.
📱 Social Scheduler	Draft all platforms from one instruction → stage for review → schedule via Buffer.

DEVELOPER SKILLS

SKILL	WHAT IT DOES
🦠 Sentry → Codex → PR	Error alert → agent identifies fix → Codex writes patch → PR opened → Slack notification. Auto.
🚀 Deploy Monitor	Watch Railway/Render/Fly. Notify on success/failure. Auto-rollback on critical errors.
👁️ Code Review Assistant	GitHub webhook → agent reviews PRs for style/logic/security → posts inline comments.
🔧 Self-Extending Agent	User describes new workflow → agent writes the skill file → user approves. Agent builds its own tools.
🖥️ Infra Manager	SSH to server, edit NixOS/Docker configs, run rebuilds — all via chat, no laptop needed.

BUSINESS SKILLS (PREMIUM)

SKILL	WHAT IT DOES
🤝 Client Onboarding Autopilot	CRM webhook → welcome email + project folder + Slack channel + calendar invite. Full sequence in minutes.
💰 Financial Pulse	Stripe + QuickBooks. Daily revenue report, P&L summary, anomaly flags. Ask in plain language.
🏆 CRM Assistant	HubSpot/Airtable: update deal stages, enrich contacts, schedule follow-ups, draft outreach.
🌙 Overnight Business Stack	While you sleep: monitor competitors + repurpose content + audit tasks + generate next-day briefing.
🏠 Family/Household Manager	Aggregate family calendars, household inventory, coordinate schedules. One agent for the household.

👤 09 — USE CASES BY PERSONA

Every use case below has been documented by real OpenClaw community members in production. Enclave makes all of it available in 2 minutes.

🚀 Solo Founder

- 4 specialized agents: strategy, dev, marketing, business — coordinated via Telegram
- Morning MRR + GitHub PRs + priority emails before opening laptop
- Overnight: competitor scraping + market gap analysis
- Sentry → Codex → PR all autonomous
- Content pipeline on schedule while founder focuses on product

💼 Freelancer / Consultant

- New client name → research brief + project folder in workspace
- Monthly invoices via QuickBooks, auto-sent on the 1st
- Before every call: surfaces all prior history + open deliverables
- Proposal writer: research + case studies + draft in one Telegram message

📊 Knowledge Worker / Analyst

- Daily intelligence brief: industry news + SEC filings + arXiv papers
- Meeting transcript → action items → tasks → email attendees
- Drop URLs into chat → searchable knowledge base, retrievable by meaning
- 20-company earnings tracker — alerts on anything relevant

👨‍💻 Developer

- Rebuilt entire website + migrated 18 posts + moved DNS — from bed, via Telegram
- Sentry alert → fix → PR → Slack before dev sees it
- Edit NixOS/Docker configs via chat. Agent SSHs + rebuilds.
- GitHub webhook → agent reviews every new PR inline

🎬 Content Creator

- Daily YouTube digest from 50 channels → workspace notes
- One piece of content → 4 platform formats in one command
- Weekly Reddit/X pain point mining

🏢 Executive / Team Lead

- 200+ daily emails → agent triages, drafts responses, flags the 5 that matter
- Jira + Slack + GitHub → standup summary before morning meeting

→ feeds content calendar

- Newsletter engine: research + draft + schedule. Remembers previous issues.

- Monthly: Stripe MRR + product metrics → agent writes investor update
- Delegation tracker: follows up on assigned tasks automatically

🏠 Personal / Household

- Full year meal plan in Notion, shopping lists by store + aisle
- Family calendar aggregation + conflict detection
- Home inventory log: "when did I buy the air purifier?" → instant answer
- Weekly goal accountability with perfect memory of every commitment

✦ 10 — BRAND IDENTITY

Name: Enclave

A geopolitical term for a territory that is fully surrounded by, but completely distinct from, another territory — with its own sovereignty. Private. Permanent. Self-governed.

WHY IT WORKS	EXPLANATION
Privacy & Sovereignty	Impenetrable, self-governed — exactly what we promise about user data
Workspace / Place	"Your enclave" is a place you go, not a tool you use
Permanence	An enclave doesn't disappear — mirrors the never-forget memory promise
Premium feel	Uncommon enough to feel distinctive without being contrived

Domain targets	`enclave.ai` · `enclave.app` · `enclaveai.com` — pursue immediately

> **The Pitch:** "Enclave gives you a hosted AI co-worker with perfect memory, full access to your calendar, docs, and apps, and a private workspace that even we cannot see. Set up in 60 seconds."

BRAND PRINCIPLES

- **Privacy is architectural, not a policy** — "We cannot read your data" not "we will not"
- **Memory is a first-class feature** — the never-forget promise is the headline
- **Co-worker, not chatbot** — assigned work, not asked questions
- **Calm and professional** — think Notion, not Duolingo. No gamification.
- **Emerald accent, once per screen max** — never decorative

💳 11 — PRICING

STARTER

$9/mo

- 1 agent
- Telegram + web chat
- lossless-claw memory
- 5GB workspace
- 3 integrations
- 3 pre-built skills

PRO ✦

$29/mo

- 1 agent
- All 25+ channels
- Full memory stack
- 25GB workspace
- Unlimited integrations
- All core skills

BUILDER

$59/mo

- 3 agents
- Multi-agent routing
- 100GB workspace
- All skills
- Dev integrations
- Webhook access

TEAMS

$49/seat

- Min 3 seats
- Shared knowledge base

- Admin dashboard
- SSO
- Priority support

Skills add-ons: $3–$9/skill/mo

Encrypted backup: +$4/mo (any tier)

UNIT ECONOMICS

METRIC	ESTIMATE
GCP VM cost per ~10 users	~$40/mo → $4/user
Memory stack overhead	Negligible — all runs in-container
Gross margin @ $9/mo	~55%
Gross margin @ $29/mo	~80%
MRR @ 100 users (mixed)	~$2,200/mo
MRR @ 500 users (mixed)	~$11,000/mo
MRR @ 2,000 users (mixed)	~$45,000/mo

🗺️ 12 — ROADMAP

PHASE 1

Now → Month 1

- lossless-claw on all instances
- Persistent volumes in Dokploy
- Encryption keys at provision
- Stripe webhook tested
- Full signup → Telegram flow
- Basic workspace (file upload)
- Discord + Slack channels

PHASE 2

Month 2

- Qdrant sidecar per instance
- Semantic search via MCP
- Web Chat in dashboard
- Google Calendar + Gmail
- Morning Briefing + Meeting Prep
- Pro tier launch

PHASE 3

PHASE 4

PHASE 3

Month 3

- Mem0 sidecar
- Memory Dashboard UI
- Notion + Sheets + TickTick
- Knowledge Base in workspace
- Inbox Zero + Weekly Review
- Signal + iMessage

PHASE 4

Month 4–5

- Skills Marketplace (10 skills)
- Content + research skills
- HubSpot + Stripe + Airtable
- Builder tier launch
- Encrypted backup add-on
- 3rd-party skill dev program

PHASE 5

Month 5–6

- Teams tier
- Dev skills (Sentry→PR, infra)
- Voice channel (Talk Mode)
- WhatsApp + Matrix + Teams
- Memory export (portability)
- iOS/Android companion apps

PHASE 6

Month 7–12

- Enterprise tier (SSO + compliance)
- Live Canvas in dashboard
- Marketplace revenue share
- EU GDPR certification
- APAC channels (WeChat, Zalo)

⚙️ 13 — TECH STACK

LAYER	TECH
Frontend	Next.js 15 (App Router), TypeScript, Tailwind CSS, shadcn/ui
Backend	Next.js API routes, Server Actions, Prisma ORM
Platform DB	PostgreSQL — metadata only. Zero user conversation data.
Auth	NextAuth v5 (Auth.js), JWT, `AUTH_TRUST_HOST=true` in prod
Payments	Stripe Subscriptions + Webhooks
Agent Runtime	OpenClaw (1 container per user)
Memory L1	lossless-claw (SQLite DAG, verbatim recall)

Memory L2	Qdrant (self-hosted vector DB, semantic recall)
Memory L3	Mem0 (self-hosted entity extraction)
Workspace Storage	Persistent Docker volumes, AES-256 encryption
Integrations	MCP (Model Context Protocol) tool bridge
Skills Platform	ClawHub (1,700+ community skills) + Enclave curated catalog
i18n	next-intl — EN + FR
Provisioning	Dokploy REST API on GCP e2-standard-2 (8GB RAM)

🏗️ 14 — INFRA & OPS

THING	WHERE
Repo	`github.com/mtldev-dotcom/clawhost`
Local dev	`localhost:3000` + GCP PostgreSQL
Production	`https://nestai.nickybruno.com`
Dokploy Panel	`http://35.202.32.236:3000`
GCP DB	`35.202.32.236:5432/nestai`
GCP VM	e2-standard-2 (8GB RAM) — upgraded 2026-03-28
DNS (user subdomains)	`*.nickybruno.com → 35.202.32.236`
DNS (Dokploy)	`dok.nestai.nickybruno.com → 35.202.32.236`

KEY FILES

FILE	WHAT IT IS
`src/lib/dokploy.ts`	Single source for all Dokploy API calls
`src/lib/stripe.ts`	Single source for Stripe calls

`src/lib/env.ts`	Zod env validation
`src/middleware.ts`	Must be in `src/` — not project root
`docs/AGENT_*.md`	Sub-agent specs (DB, Auth, Stripe, Provisioner, Dashboard, Skills)

⚠️ 15 — WATCH OUT

CRITICAL ISSUES

- **lossless-claw needs PR #22201** — not merged upstream yet. Fork + patch. Top priority before memory marketing.
- **Persistent volumes** — must be mounted in Dokploy before lossless-claw ships. SQLite can't live in ephemeral container storage.
- **Stripe webhook** — not tested end-to-end in production yet. Must validate before public launch.
- **DNS updates needed** — `*.nickybruno.com` + `dok.nestai.nickybruno.com` must point to new IP `35.202.32.236`

KNOWN GOTCHAS

- Next.js 15 + next-auth has prerender bug on error pages — build warnings are OK
- `AUTH_TRUST_HOST=true` required for NextAuth in production
- Middleware **must** be in `src/middleware.ts` not project root
- Dokploy inline compose needs `sourceType: "raw"`
- OpenClaw uses `TELEGRAM_BOT_TOKEN` + `OPENAI_API_KEY` env vars — not OPENCLAW_*
- Brand: emerald accent once per screen max — never decorative
- Layouts: Root layout has NO nav. Public pages use PublicNav. Dashboard uses DashboardHeader.

RISK TABLE

RISK	LEVEL	MITIGATION
lossless-claw PR dependency	Medium	Fork + apply patch. Contribute to

		PR.
Compute costs at scale	Medium	Optimize container density. Hetzner for EU.
OpenClaw upstream breaks	Low-Med	Pin versions per instance. Staging before rollout.
User data loss (container)	Medium	Persistent volumes + nightly backup option.
Privacy claim creates target	Low	Architecture is auditable. GDPR legal opinion before EU.
LLM provider outages	Low	Model failover + OpenRouter fallback built into OpenClaw.

ONLY PAUSE FOR NICK IF

- A Stripe product/price ID needs manual creation in the dashboard
- Dokploy API key not in `.env.local`
- OpenClaw Docker image name unclear (default: `ghcr.io/openclaw/openclaw:latest`)